

single table, and how many tables you need. A database is only as efficient as the table structure.

Each table needs a “primary key.” In plain English, this simply means a unique value—like a serial number or ID—that is not likely to be repeated in the table. So select a field as the primary key, or if you are not too sure, let Access do it for you. Take care with the data types of the fields. You get a short description of each data type, so it’s very unlikely that you’ll use Text as the data type if your field is Date. Browse through the options and select what seems to be the best fit. Give a name for your table and save it. This appears in the Database window.

Click on the newly-created table and enter your data. If you have done your spadework right, the data will fit nicely into their places. If you goofed up, sweat not: go to the Database window, right-click on the table, select Design view from the menu, and tweak the field types. If you need to add or delete columns, you can do so from inside the table.

You can edit the data in tables just like you would in an Excel worksheet

Student info : Table					
	Student ID	Name	City	Email	Phone number
▶	1	Vishal	Mumbai	vishal@xxx.com	2305687
	2	Madhur	Delhi	madhur@xxx.com	
	3	Radha	Mumbai	radha@xxx.com	3405676
	4	John	Bangalore	john@xxx.com	
	5	Amrit	Chennai	amrit@xxx.com	3459374
	6	Mayur	Ahmedabad	mayur@xxx.com	5607859
	7	Priya	Delhi	priya@xxx.com	2365437
	8	Meera	Mumbai	meera@xxx.com	
	9	Akash	Delhi	akash@xxx.com	
	10	Ravi	Mumbai	ravi@xxx.com	2313547
*	(AutoNumber)				

Record: 1 of 10

The student info table, one of the two tables used in our sample database

Fees : Table				
	Student ID	Subject taken	Month Joined	Fees payable
▶	1	Physics, Chemi	January	500
	2	Maths	February	250
	3	Physics, Chemi	December	750
	4	English, Hindi	January	500
	5	Maths, Chemist	February	500
	6	Sanskrit	February	250
	7	English	February	250
	8	Maths, Chemist	January	500
	9	Hindi	February	250
	10	Physics, Chemi	February	1000
*	(AutoNumber)			0

The second table used by us, the Fees table

You can add data to your table—or delete data—simply by choosing to edit it.

Queries

Queries are used to extract data from databases. In Access, queries are of five types—select queries, action queries, parameter queries, crosstab queries, and SQL queries. Most users

deal with select queries—those that retrieve required data from your tables and display them in spreadsheet view. Action queries do jobs like deleting or updating records. Parameter queries are interactive: a single query can give different results, depending on your parameter. Crosstab queries are used for restructuring data and doing operations like addition, calculating averages, etc. SQL queries are written in SQL (Structural Query Language) and are the most powerful of them all. In fact, all the other types of queries are converted by Access into the equivalent SQL forms and then executed. However, SQL queries are not suitable for the lay user because its grammar and syntax is complex.

From the Database window, click on the Query tab and select New. You get five options—Create query in Design view, and create query using the Wizards (for Simple Query, Crosstab Query, Duplicate Query, and Unmatched Query). Parameter queries and SQL queries can be created using the Design view. For our student database, we have used the Design view to create queries, though the job can be done using the Wizard too.

Click on Design view. Select the tables that contain the data for your query. In fact, you can also create sub-queries by selecting the queries you created earlier. Select the requisite data fields from the tables, add a sorting order if necessary, and save and name your query. That’s all—click on the created query in the Database window and view the results. As an example, we have used both the Fees and the Student Info table and selected appropriate fields to create a query that relates student ID, name, subjects taken, and month joined.

Parameter queries can be created by adding an expression typed in square brackets with a prompt in the Criteria cell of the Field to be used

Query1 : Select Query				
	Fees	Student info		
	* Student ID	* Student ID		
	Subject taken	Name		
	Month Joined	City		
	Fees payable	Email		
Field:	Student ID	Name	Subject taken	Month Joined
Table:	Fees	Student info	Fees	Fees
Sort:				
Show:	☒	☒	☒	☒
Criteria:				
or:				

This query was created in the Design view...

Student city : Select Query				
	Student ID	Name	Subject taken	Month Joined
▶	1	Vishal	Physics, Chemistry	January
	2	Madhur	Maths	February
	3	Radha	Physics, Chemistry, Biology	December
	4	John	English, Hindi	January
	5	Amrit	Maths, Chemistry	February
	6	Mayur	Sanskrit	February
	7	Priya	English	February
	8	Meera	Maths, Chemistry	January
	9	Akash	Hindi	February
	10	Ravi	Physics, Chemistry, Maths, Hindi	February
*	(AutoNumber)			

Record: 1 of 10

And here are the results of the query